

QUICK REFERENCE QR-001

WAFT: THE EVOLUTIONARY CODE LABORATORY

Essential Information - One Pager

PUBLIC

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WAFT: THE EVOLUTIONARY CODE LABORATORY

Don't just build agents. Breed them. Python framework for directed evolution of self-modifying AI agents. Scientific instrument for studying **physics of artificial cognition**.

THE THREE PILLARS

1. The Substrate (Code as DNA)

Genome ID: SHA-256 hash | **Mutations:** Code/config/prompt changes | **Evolution:** Hot-swap genomes

2. The Physics (Scint System)

Reality Fracture Detection = natural selection. Errors: **SYNTAX_TEAR** (formatting), **LOGIC_FRACTURE** (math), **SAFETY_VOID** (harmful), **HALLUCINATION** (fabricated).

Fitness = (Stability × 0.4) + (Efficiency × 0.3) + (Safety × 0.3) | < 0.5 = DEATH

3. The Flight Recorder

Telemetry: Genome ID, Parent ID, Generation, Event Type, Context, Fitness. Enables **Family Tree** reconstruction.

PUBLIC

QUICK START

```
uv tool install waft
waft new my_laboratory && cd my_laboratory
waft verify
```

EVOLUTIONARY CYCLE

Spawn (variants) → **Evaluate** (fitness) → **Evolve** (adopt best) → **Record** (track all)

KEY FEATURES

Self-Modification | **Evolutionary Tracking** | **Fitness Testing** | **Scientific Data** | **Gamification** | **Epistemic Tracking**

ESSENTIAL COMMANDS

Project: waft new <project> | waft verify | waft status

Evolution: waft spawn --agent <name> | waft eval --agent <name> | waft evolve --agent <name>

Docs: waft-docs | waft-status

Goal: Observe "God-Head" agent emerge from thousands of generations. Produces data for "The Physics of Artificial Cognition" research.

Version: 0.5.0 | **Python:** 3.10+ | **Package Manager:** uv